Industry 4.0 Glossary

What Is Model Training?

Model training is the process of training an ML algorithm with adequate training data to demonstrate correlation between the outcome and the influencing variables.

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A machine learning training model is a process in which a machine learning (ML) algorithm is fed with sufficient training data to learn from.

ML models can be trained to benefit manufacturing processes in several ways. The ability of ML models to process large volumes of data can help manufacturers identify anomalies and test correlations while searching for patterns across the data feed. It can equip manufacturers with predictive maintenance capabilities and minimize planned and unplanned downtime.

What Is Model Training In Machine Learning?

A training model is a dataset that is used to train an ML algorithm. It consists of the sample output data and the corresponding sets of input data that have an influence on the output. The training model is used to run the input data through the algorithm to correlate the processed output against the sample output. The result from this correlation is used to modify the model.

This iterative process is called "model fitting". The accuracy of the training dataset or the validation dataset is critical for the precision of the model.

Model training in machine language is the process of feeding an ML algorithm with data to help identify and learn good values for all attributes involved. There are several types of machine learning models, of which the most common ones are supervised and unsupervised learning.

Supervised learning is possible when the training data contains both the input and output values. Each set of data that has the inputs and the expected output is called a supervisory signal. The training is done based on the deviation of the processed result from the documented result when the inputs are fed into the model.

Unsupervised learning involves determining patterns in the data. Additional data is then used to fit patterns or clusters. This is also an iterative process that improves the accuracy based on the correlation to the expected patterns or clusters. There is no reference output dataset in this method.

Creating A Model In Machine Learning

There are 7 primary steps involved in creating a machine learning model. Here is a brief summarized overview of each of these steps:

1. **Defining The Problem**
   Defining the problem statement is the first step towards identifying what an ML model should achieve. This step also enables recognizing the appropriate inputs and their respective outputs; Questions like "what is the main objective?", "what is the input data?" and "what is the model trying to predict?" must be answered at this stage.
2. **Data Collection**
   After defining the problem statement, it is necessary to investigate and gather data that can be used to feed the machine. This is an important stage in the process of creating an ML model because the quantity and quality of the data used will decide how effective the model is going to be. Data can be gathered from pre-existing databases or can be built from the scratch
3. **Preparing The Data**
   The data preparation stage is when data is profiled, formatted and structured as needed to make it ready for training the model. This is the stage where the appropriate characteristics and attributes of data are selected. This stage is likely to have a direct impact on the execution time and results. This is also at the stage where data is categorized into two groups – one for training the ML model and the other for evaluating the model. Pre-processing of data by normalizing, eliminating duplicates and making error corrections is also carried out at this stage.
4. **Assigning Appropriate Model / Protocols**
   Picking and assigning a model or protocol has to be done according to the objective that the ML model aims to achieve. There are several models to pick from, like linear regression, k-means and bayesian. The choice of models largely depends on the type of data that is being used. For instance, image processing convolutional neural networks would be the ideal pick and k-means would work best for segmentation.
5. **Training The Machine Model Or "The Model Training"**
   This is the stage where the ML algorithm is trained by feeding datasets. This is the stage where the learning takes place. Consistent training can significantly improve the prediction rate of the ML model. The weights of the model must be initialized randomly. This way the algorithm will learn to adjust the weights accordingly.
6. **Evaluating And Defining Measure Of Success**
   The machine model will have to be tested against the "validation dataset". This helps assess the accuracy of the model. Identifying the measures of success based on what the model is intended to achieve is critical for justifying correlation.
7. **Parameter Tuning**
   Selecting the correct parameter that will be modified to influence the ML model is key to attaining accurate correlation. The set of parameters that are selected based on their influence on the model architecture are called hyperparameters. The process of identifying the hyperparameters by tuning the model is called parameter tuning. The parameters for correlation should be clearly defined in a manner in which the point of diminishing returns for validation is as close to 100% accuracy as possible.

How Long Does It Take To Train A Machine Learning Model?

There is no definite period or a prefixed set of iterations for training an ML model. The factors influencing the duration of training could be the quality of training data, proper definition of the measures of success and complexity of model selection. Factors like method of training, allocation of weights and complexity of the model also play an important role. Other factors that are extraneous to the data or the models like computing power and skilled resources can also have an impact on the training duration. There is always a scope for optimizing training a model as the number of parameters influencing its duration are too many.

How Can Manufacturers Use Machine Learning?

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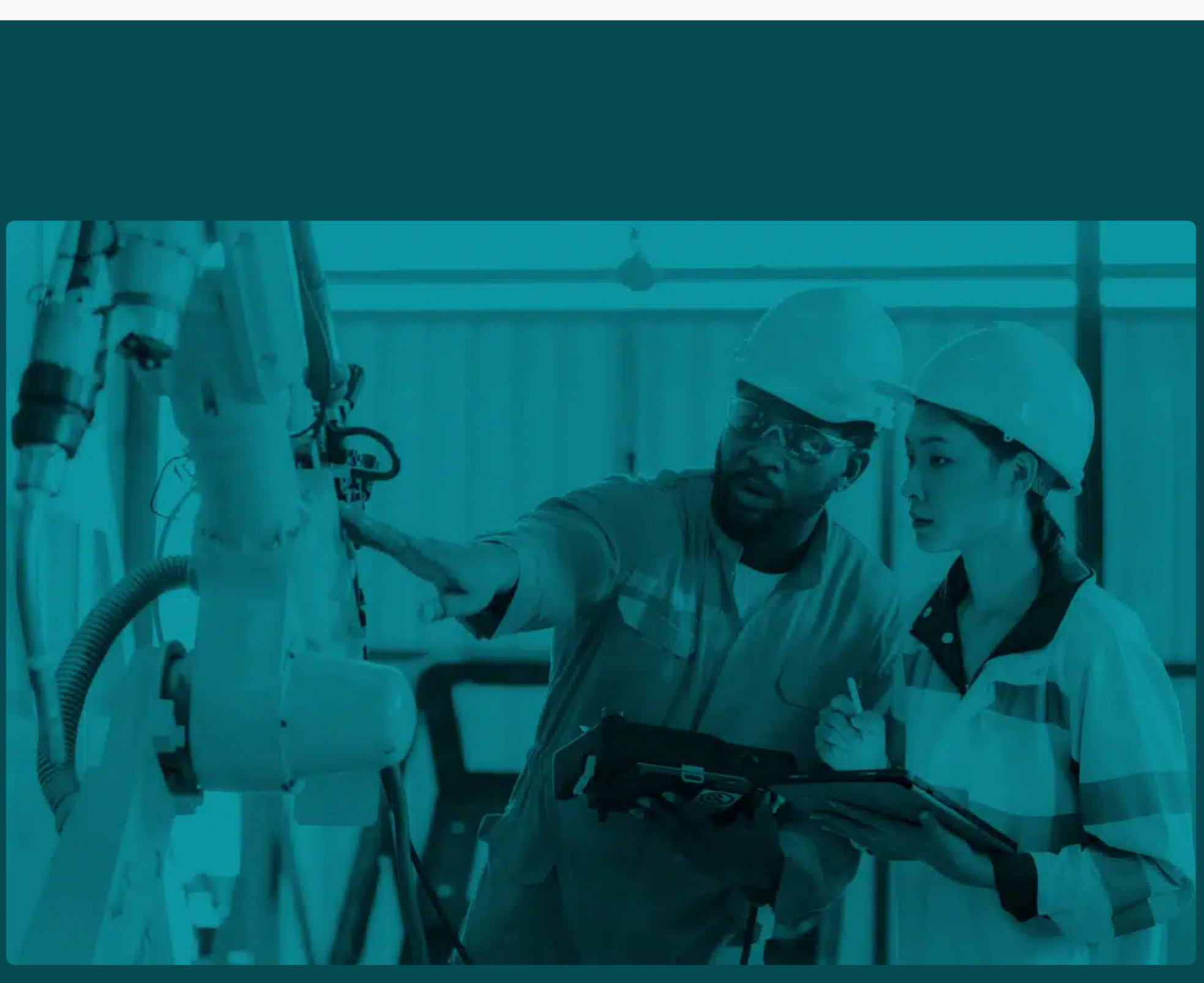

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